CMPE293 Homework #4
Answers to Selected Problems

4.4

a) $$\frac{Y(s)}{W(s)} = \frac{-1500s}{s^2 + (60 + 600Kp)s + 600K_I}$$

b) Kp = 1/10, K_I = 12

4.18

b) K must be POSITIVE to stabilize the system

4.26

a) $$\frac{Y(s)}{R(s)} = \frac{10(K_P s + K_I)}{s^3 + s^2 + (20 + 10Kp)s + 10K_I}$$

b) $$\frac{Y(s)}{W(s)} = \frac{10s}{s^3 + s^2 + (20 + 10Kp)s + 10K_I}$$

c) $K_I > 0$ and $K_P > (K_I - 2)$

d) System is TYPE 1
Error Constant $K_v = K_I/2$

e) System is TYPE 1
Error Constant $K_v = -K_I$

4.35

a) $$\frac{Y(s)}{R(s)} = \frac{A}{s^2 + (2 + AK_t)s + A}$$

c) $$\frac{2 + AK_t}{A} \le 0.10$$

e) $$e_{SS} = \frac{2 + AK_t}{A}$$

f) $K_t = 0.181$ will get your poles on the "boundary" but does not satisfy SS error condition. Reduce K_t in MATLAB to satisfy SS error BUT then the other specs become violated. Therefore, there is no K_t that will satisfy ALL 3 conditions for A=32.

g) $t_s \cong 1.18s$ (using $K_t = 0.181$ and A=32)